

6.8 FERMI-DIRAC DISTRIBUTION

We know that in a metal, in the absence of an electric field, free electrons move at random, similar to the behaviour of the electron gas. Since electrons are indistinguishable particles, they are known as *fermions* or *Fermi particles*. Hence, such an electron gas obeys Fermi-Dirac statistics or distribution. The Fermi-Dirac distribution describes the behaviour of free electron gas, taking into account the quantum theory and Pauli's exclusion principle.

As shown in Fig. 6.15, according to quantum theory, the electrons will have discrete energy states or levels and the occupation or distribution of electrons among these energy levels will be governed by Pauli's exclusion principle, i.e., not more than two electrons can occupy the same energy level. At absolute zero temperature, two electrons occupy the ground state and two into each state of next higher energy levels. The highest energy level occupied by electrons at absolute zero is known as Fermi energy level which divides or separates the occupied states from the unoccupied states. The energy of the Fermi level is E_F . The probability of the occupation of an energy level E by an electron at temperature T is given by

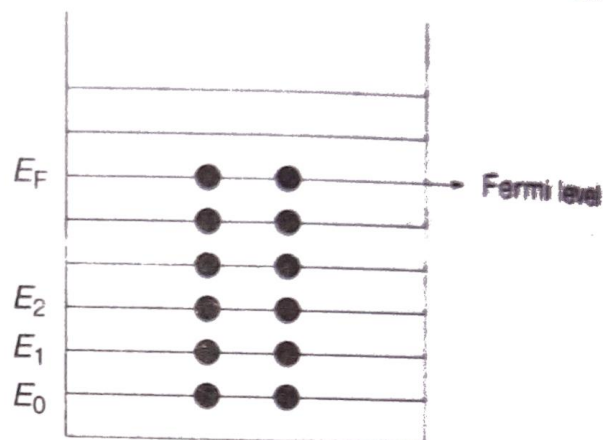


Figure 6.15 Occupation of energy levels at 0 K

$$F(E) = \frac{1}{1 + e^{(E - E_F)/K_B T}}$$

where $F(E)$ is the Fermi-Dirac distribution function and K_B is the Boltzmann's constant.

6.8.1 Effect of Temperature on Fermi-Dirac Distribution Function

Figure 6.16 shows plots of the distribution function against energy at different temperatures. The special features of the plots are discussed below.

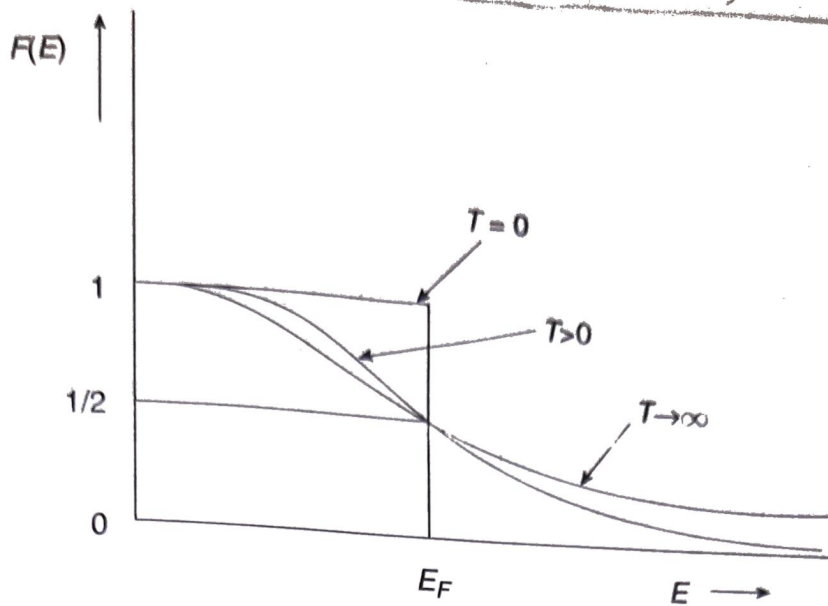


Figure 6.16 Variation of Fermi distribution function with energy

Case I – At temperature $T = 0\text{K}$

- (a) For $E < E_F$, $\frac{E - E_F}{K_B T} \rightarrow -\infty$ and $F(E) = 1$
- (b) For $E > E_F$, $\frac{E - E_F}{K_B T} \rightarrow +\infty$ and $F(E) = \frac{1}{\infty} = 0$

Thus, at absolute zero temperature $F(E)$ is a step function. All the states with energies up to E_F are filled with electrons and the states with energies greater than E_F are empty. Thus the Fermi level may be defined as the topmost filled level at absolute temperature.

Case II – At temperature $T > 0\text{K}$

- (a) For $E < E_F$, $\frac{E - E_F}{K_B T}$ is negative and

$$F(E) \cong 1 \text{ for } E \ll E_F$$

- (b) For $E > E_F$, $\frac{E - E_F}{K_B T}$ is positive and

$$F(E) \cong 0 \text{ for } E \gg E_F$$

- (c) For $E = E_F$, $\frac{E - E_F}{K_B T} = 0$ and $F(E) = 1/2$.

Thus, at finite temperature the probability of occupancy of the Fermi level is $1/2$. The Fermi function when multiplied with the density of states gives the number of states per unit volume actually occupied by the electrons.